

Project 5:

Natural Language Processing (NLP)

Application on Wikipedia Articles:

A Retrospective

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Agenda

- Project Overview
- Part 1: Similarity Scores
- Part 2: K-Nearest Neighbor
- Part 3: Interactive Code
- Reflection

Project Overview

- Data Cleaning
- NLP application using:
 - ♦ TF-IDF
 - ♦ Vectorization
 - ♦ K Nearest Neighbors
- Output similar persons
- Visualize results
- Interactive notebook

	URI	name	text
0	<http://dbpedia.org/resource/Digby_Morrell>	Digby Morrell	digby morrell born 10 october 1979 is a former...
1	<http://dbpedia.org/resource/Alfred_J._Lewy>	Alfred J. Lewy	alfred j lewy aka sandy lewy graduated from un...
2	<http://dbpedia.org/resource/Harpdog_Brown>	Harpdog Brown	harpdog brown is a singer and harmonica player...
3	<http://dbpedia.org/resource/Franz_Rottensteiner>	Franz Rottensteiner	franz rottensteiner born in waidmannsfeld lowe...
4	<http://dbpedia.org/resource/G-Enka>	G-Enka	henry krivits born 30 december 1974 in tallinn ...

Above: Similar people output using TF-IDF

42,786
rows

505
.iloc select

Part 1: Similarity Scores

TfidfVectorizer

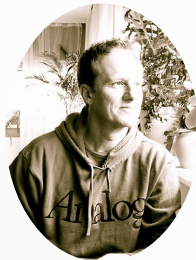
TextBlob

distances

NearestNeighbors

sentiment

indices



Filippo Gaetani [505]

0.10

sentiment

0.31

subject

Andrew Weiss (guitarist) has a score of:
Sentiment(polarity=0.09143385070214338, subjectivity=0.3472025129342203)

Bob Both has a score of:
Sentiment(polarity=0.1402121781727045, subjectivity=0.3920146958304852)

Butch Taylor has a score of:
Sentiment(polarity=0.12363636363636361, subjectivity=0.33499999999999996)

Dan Siegel (musician) has a score of:
Sentiment(polarity=0.18787878787878787, subjectivity=0.40629734848484844)

Filippo Gaetani has a score of:
Sentiment(polarity=0.10084670231729057, subjectivity=0.3161764705882353)

James Hoover has a score of:
Sentiment(polarity=0.00535353535353535, subjectivity=0.3327777777777778)

John Burns (audio engineer) has a score of:
Sentiment(polarity=0.10402837902837903, subjectivity=0.3716625966625967)

Oliver Hartmann has a score of:
Sentiment(polarity=0.10266955266955267, subjectivity=0.3302669552669553)

Paul Goldberg (musician) has a score of:
Sentiment(polarity=0.24389545454545454, subjectivity=0.4804606060606061)

Simone Sello has a score of:
Sentiment(polarity=0.10312049062049063, subjectivity=0.2626984126984127)

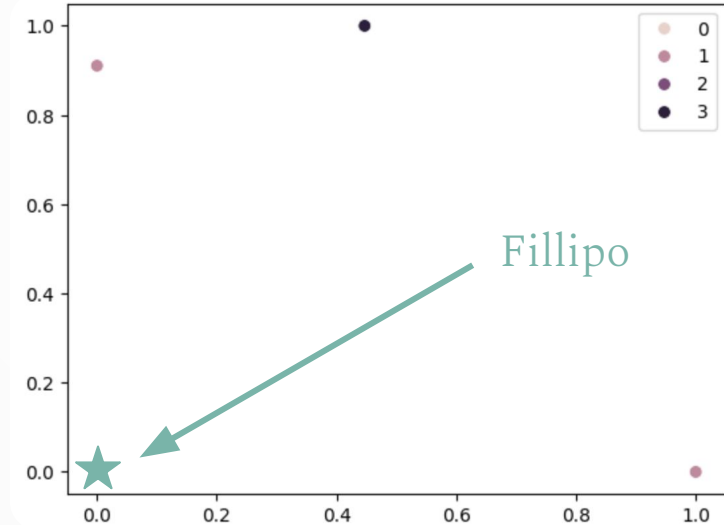
Steff Fontaine has a score of:
Sentiment(polarity=-0.08205128205128204, subjectivity=0.4051282051282052)

Above: Similar musicians to Filippo
based on TF-IDF and KNN scoring

Part 2: K-Nearest Neighbor

	name	wiki_text	sentiment_score	sentiment_score_perc	rank
3	Dan Siegel (musician)	dan siegel born in seattle, washington is an a...	0.128525	12.85	1
1	Bob Both	bob both born 1952 is an american recording en...	0.125095	12.51	2
2	Butch Taylor	butch taylor born clarence francis taylor on a...	0.122432	12.24	3
6	Oliver Hartmann	oliver hartmann born 28 june 1970 in rüsselshe...	0.109807	10.98	4
5	John Burns (audio engineer)	john burns is a british recording engineer bes...	0.098954	9.90	5
0	Andrew Weiss (guitarist)	andrew weiss is an american musician, composer...	0.090123	9.01	6

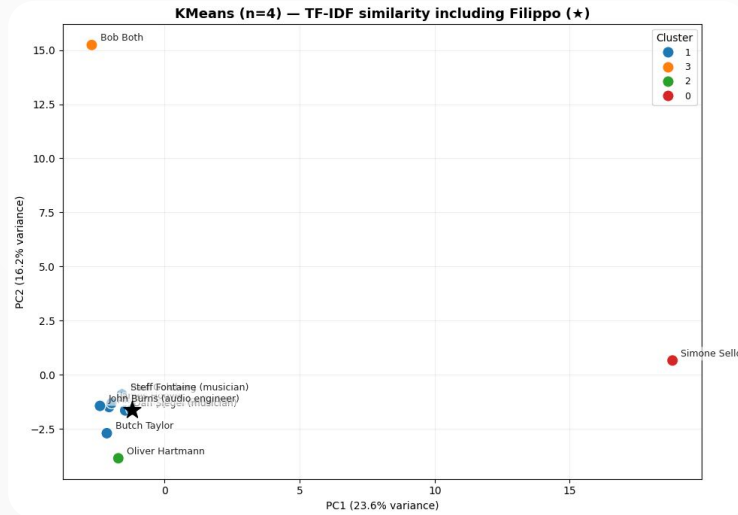
Above: Ranked results on similar persons compared to Filippo using sentiment scoring



Above: 4 Clusters from KMeans approach on similarity to Filippo (bottom left)

Part 3: Interactive Code

→ [\[Notebook\]](#)



Above: Clustered PCA results using TF-IDF and KMeans... done with Ecosia AI

Wikipedia Nearness Explorer

Person:

Find 10 Closest

10 closest individuals to: Oliver Hartmann

	Name	Cosine Similarity
1	Butch Taylor	0.1133
2	Simone Sello	0.0921
3	Andrew Weiss (guitarist)	0.0798
4	Dan Siegel (musician)	0.0792
5	Filippo Gaetani	0.0640
6	Bob Both	0.0636
7	John Burns (audio engineer)	0.0580
8	James Hoover	0.0430
9	Paul Goldberg (musician)	0.0000
10	Steff Fontaine	0.0000

Above: Interactive module on finding similar persons using built-in Wikipedia

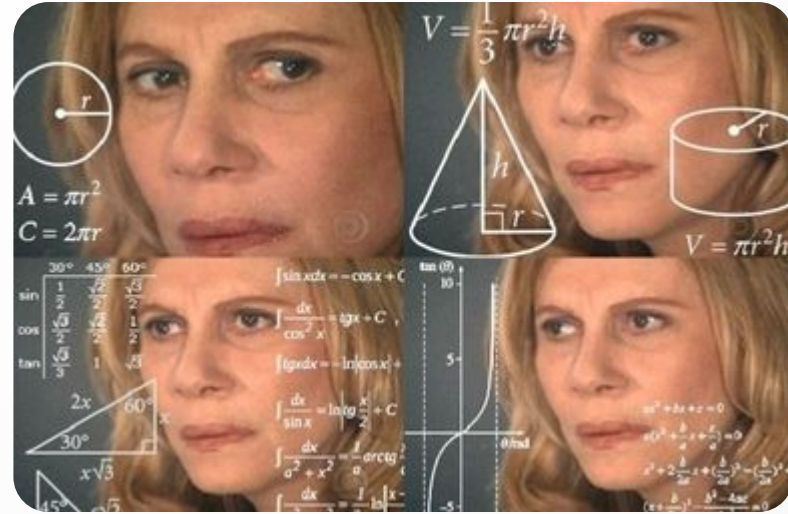
Reflection

→ Future Development:

- Develop try/except logic
- Retry DBpedia output for interactive code section

→ Challenges:

- Understanding backend code in DBpedia
- More complex data cleaning
- Review lectures in-depth



Above: How I felt during this entire project

Thank you!

Questions?

[\[View Github\]](#)